SS-MAE: Spatial–Spectral Masked Autoencoder for Multisource Remote Sensing Image Classification

Junyan Lin, Feng Gao, *Member, IEEE*, Xiaochen Shi, Junyu Dong, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Masked image modeling (MIM) is a highly popular and effective self-supervised learning method for image understanding. The existing MIM-based methods mostly focus on spatial feature modeling, neglecting spectral feature modeling. Meanwhile, the existing MIM-based methods use Transformer for feature extraction, and some local or high-frequency information may get lost. To this end, we propose a spatial–spectral masked autoencoder (SS-MAE) for hyperspectral image (HSI) and light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data joint classification. Specifically, SS-MAE consists of a spatialwise branch and a spectralwise branch. The spatialwise branch masks random patches and reconstructs missing pixels, while the spectralwise branch masks random spectral channels and reconstructs missing channels. Our SS-MAE fully exploits the spatial and spectral representations of the input data. Furthermore, to complement local features in the training stage, we add two lightweight convolutional nerual networks (CNNs) for feature extraction. Both global and local features are taken into account for feature modeling. To demonstrate the effectiveness of the proposed SS-MAE, we conduct extensive experiments on three publicly available datasets. Extensive experiments on three multisource datasets verify the superiority of our SS-MAE compared with several state-of-the-art baselines. The source codes are available at https://github.com/summitgao/SS-MAE.**

***Index Terms*— Deep learning, hyperspectral image (HSI), masked autoencoder (MAE), multisource data.**

I. Introduction

With the advancement of Earth observation technologies and satellite sensor platforms, significant amounts of remote sensing data have been obtained for a variety of purposes, such as vegetation mapping [1], coastal wetland classification [2], object detection [3], change detection [4], land cover classification [5], and so on. Among these applications, land cover classification is fundamental to many operational mapping and reporting natural resource management programs.

In recent decades, substantial efforts have been made to develop effective classifiers for remote sensing data classification using hyperspectral images (HSIs) [6], [7], [8] or polarimetric synthetic aperture radar (PolSAR) images [9], [10], [11]. Most of these methods use deep neural networks for remote sensing data classification. Multiscale feature fusion [4], dual-branch network [12], hypergraph convolution [13], [14], Transformer-like network [15], attention multihop graph [16], and neural architecture search [17] have been used for hyperspectral or SAR data feature extraction. Although impressive results have been achieved, single modality classification can be problematic in some cases. For example, it is challenging to accurately identify different classes made up of the same material using an HSI alone (such as roof and cement pavement), as they have similar spectrum corresponds. At the same time, height information from light detection and ranging (LiDAR) data can make up for the limitation. Furthermore, synthetic aperture radar (SAR) sensor possesses the inherent capability of conducting observations around the clock and under any weather conditions. This feature significantly compensates for the shortcomings of hyperspectral sensors during unfavorable weather conditions [18]. Therefore, in this article, we mainly focus on developing a robust land cover classification method via hyperspectral and LiDAR/SAR remote sensing image fusion.

Many decision-level fusion [19], [20], [21] and feature-level fusion [22], [23], [24] methods have been devoted for multisource remote sensing image classification and achieved promising performance. However, these methods commonly employ supervised training strategy, which requires substantial amounts of labeled data, making it a labor-intensive process. Afterward, unsupervised methods have been developed to alleviate the data annotation problem in multisource remote sensing image classification [25], [26]. It can directly learn general feature representations from unlabeled data. Recently, masked image modeling (MIM) has demonstrated great potential in unsupervised visual representation learning [27], [28], [29]. It learns universal feature representations from unlabeled data by reconstructing masked image patches, which can be used to achieve powerful performance in downstream tasks. However, MIM has rarely been considered in multisource remote sensing image classification, and in this article, we focus on developing a robust multisource image classification method based on MIM.

There are two challenges associated with the development of MIM-based multisource remote sensing image classification methods.

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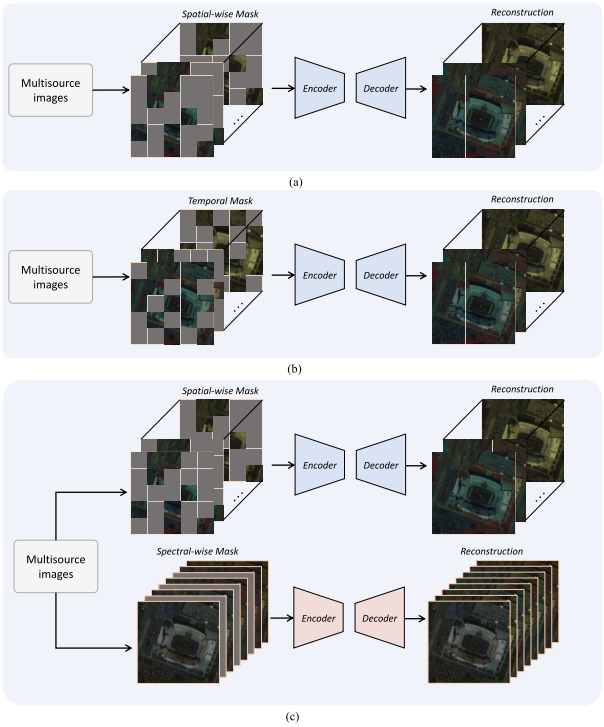


Fig. 1. Comparisons of the traditional MIMs with the proposed SS-MAE. (a) MAE for multisource data classification. (b) SatMAE for multisource data classification (c) Proposed SS-MAE.

1) *Spectral Feature Modeling:* As shown in Fig. 1, the existing MIM techniques that can be directly employed for multisource remote sensing image classification can be divided into two categories. The first category is spatialwise masking employed by masked autoencoder (MAE) [28], as illustrated in Fig. 1(a), which masks random patches in the spatial domain and ignores the spectral information. This scheme does not fully explore the spectral information, leading to limited classification performance improvement. The second category is temporal masking employed by SatMAE [30], which takes spectral information into account, as illustrated in Fig. 1(b). However, the way of masking random patches band by band may lead to information leakage, thus reducing the difficulty of reconstruction. Therefore, how to effectively integrate spectral features for MIM-based methods poses the main challenge for us.
2) *Local and Global Feature Extraction:* Transformer is commonly used for MIM, and it tends to pay more attention on global feature dependencies via long-range interactions. However, some local features or high-frequency information (such as edge and texture) may get lost. Consequently, how to simultaneously modeling the local and global features in Transformer-like network is a tough challenge.

To tackle these challenges, we present a spatial–spectral masked autoencoder (SS-MAE) for HSI and LiDAR/SAR data joint classification. Specifically, we adopt a new MIM technique, as illustrated in Fig. 1, which comprises spatialwise branch and spectralwise branch. The spatialwise branch masks random patches and reconstructs missing pixels, while the spectralwise branch masks random spectral channels and reconstructs missing channels. Hence, the proposed SS-MAE fully exploits the spatial and spectral feature representations of the input multisource data. Furthermore, to complement local features in the training stage, we add two lightweight CNNs for feature extraction. Both global and local features are used in feature modeling. To demonstrate the effectiveness of the proposed SS-MAE, we conduct extensive experiments on three publicly available datasets. The result shows that the proposed SS-MAE outperforms several state-of-the-art methods.

Our main contributions can be summarized as follows.

1) We propose SS-MAE, which performs both spatial and spectral reconstructions for feature representation learning. To the best of our knowledge, this is the first MIM work for multisource remote sensing image classification.
2) We propose a Transformer and CNN hybrid network for global and local feature modeling. Dual-attention Transformer is employed for both spectral and spatial feature pretraining. Moreover, lightweight CNNs are used to capture local features.
3) Extensive experiments on three benchmark datasets reveal that our proposed method outperforms the state-of-the-art methods. As a byproduct, we will release the codes of SS-MAE to benefit other researchers.

The remainder of this article is organized as follows. In Section II, we review closely related multisource remote sensing image classification methods and self-supervised methods. The details of SS-MAE are described in Section III. Experiments on three multisource datasets are presented in Section IV. The conclusions are drawn in Section V.

II. Related Works

A. Supervised Multisource Remote Sensing Data Classification

There are two types of supervised multisource classification methods: decision fusion [19], [20], [21] and feature fusion [22], [23], [24]. Specifically, decision-fusion methods usually use several classifiers separately and then combine the classifications together to form the final classification result. Liao et al. [19] adopted four support vector machines (SVMs) as the classifier for spectral, spatial, elevation, and graph fused features and combined the classification results by weighted summation. The combination weights are determined by the classification accuracy of each source. Ge et al. [31] gained extinction profile and local binary pattern features from both sources and classified them by collaborative representation-based classifier with Tikhonov regularization. With the popularity of deep learning, feature-fusion methods have been used more frequently. Most feature-fusion methods employ dual-stream networks to extract features from multisource remote sensing images and then converge the extracted features through various interaction approaches. Li et al. [24] presented a dual-channel framework, amalgamating spatial, spectral, and multiscale attention mechanisms. This model was devised to extract features and classify

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multisource remote sensing data. Gao et al. [32] designed a depthwise cross-attention module to extract not only self-correlation but also cross correlation from diverse multisource data. Wang et al. [33] proposed adaptive mutual-learning-based multimodal network (AM^{3}Net), which includes an involution operator, a spectral–spatial mutual-guided module, and a spectral–spatial mutual-guided module.

Although these supervised learning-based methods have achieved excellent performance, their performance is highly dependent on the large-scale labeled training samples. Correspondingly, self-supervised learning methods that do not require many training samples may overcome this limitation.

B. Self-Supervised Learning

Self-supervised learning can process unlabeled data to obtain robust representations that can facilitate downstream learning tasks. The existing self-supervised learning methods can be divided into contrastive-based [34], [35], [36], [37] and reconstruction-based [27], [28], [30], [38].

Contrastive-based methods learn feature representations by minimizing the distance between positive samples while maximizing the distance between negative samples. Simple framework for contrastive learning of visual representations (SimCLR) [35] harnesses the power of contrastive loss to learn visual representations. It achieves this by maximizing agreement between differently augmented views of the identical data sample. Liu et al. [37] demonstrated the effectiveness of using contrast learning in remote sensing data classification. Also, contrastive-based methods are beginning to be used in the multimodal remote sensing image classification. Feng et al. [39] use joint intra- and cross-modal contrastive learning to mine multimodal feature representations during pretraining. They design a simple yet effective hybrid cross-modal fusion module in the fine-tuning stage to better compactly integrate complementary information across these modalities for ground object classification.

Reconstruction-based methods learn to predict plausible content for the masked regions based on the information provided by the visible regions. The training is commonly guided by a loss function that measures the difference between the predicted completion and the ground-truth content in the masked regions. He et al. [28] proposed a simple yet effective method to pretrain large vision models by masking patches of an image and using an autoencoder to predict the masked patches. Cong et al. [30] presented a reconstruction-based pretraining framework for multispectral satellite imagery, called SatMAE. Although yielding excellent performance, the length of token sequences in SatMAE becomes excessively long when dealing with high-dimensional spectral data. This could potentially lead to the challenge of consuming excessive computational resources. Furthermore, the proposed SS-MAE is based on a Transformer and CNN hybrid network, which results in enhanced computational efficiency compared with SatMAE.

To this end, we presented a two-branch autoencoder framework for spatial and spectral feature modeling, respectively. Considering that the model is reconstructing images that are masked along an entire spatial or spectral dimension, such a framework is preferable for HSI feature modeling. To the best of our knowledge, we are the first to use MIM for multisource remote sensing classification tasks.

C. Vision Transformers

Vaswani et al. [40] introduced the Transformer model, which relies exclusively on attention mechanisms, and it has demonstrated exceptional performance in the field of natural language processing. Recently, Transformer-based architecture quickly becomes popular in computer vision [41], [42], [43], [44] and remote sensing [15], [45], [46], [47]. In multimodal remote sensing image classification, Transformer-like architectures have been explored recently. Zhang et al. [48] present a multimodal Transformer network (MTNet) that captures both the specific and shared features of HSI and LiDAR data. Roy et al. [49] introduce a multimodal fusion Transformer network, which uses cross-patch attention for remote sensing data classification. The cross-patch attention employs complementary information from LiDAR in the Transformer encoder to achieve better generalization.

Motivated by the success of vision Transformers (ViTs), most of the recent MIM methods utilize the Transformer for feature extraction. Transformer-based methods focus on learning long-range dependencies to obtain better feature representations, and they ignore some local or high-frequency information (such as textures and corners) [50]. Due to the complexity of ground objects in remote sensing images, local features are essential for discriminative representation. To this end, we add two lightweight CNN in the training stage for local feature extraction.

III. Methodology

In this section, we first describe the framework of the proposed SS-MAE and then detail the pretraining stage, dual-attention Transformer network, the training and inference stages.

A. Proposed SS-MAE Framework

The proposed SS-MAE framework is shown in Fig. 2, which consists of two parts: the pretraining stage and the training stage. For a given pair of HSI patch $X_1 \in \mathbb{R}^{C_1\times P\times P}$ and SAR/LiDAR patch $X_2 \in \mathbb{R}^{C_2\times P\times P}$ covering the same region, they are first concatenated to $T \in \mathbb{R}^{(C_1+C_2)\times P\times P}$, and then, two different masking strategies are applied to T, i.e., spatialwise masking and spectralwise masking, to generate two different masked images, T^M_{spa} and T^M_{spe}. Afterward, T^M_{spa} and T^M_{spe} are fed into the SS-MAE encoder–decoder to obtain the reconstructed images R_{spa} and R_{spe}, respectively. In the training stage, in addition to extracting the global representations of the two-branch MAE encoder, the two inverted residual blocks (IRBs) are used to extract the high-frequency information-rich representations of X_1 and X_2, respectively, and the final classification results are obtained by concatenating these representations with the classifier.

B. Pretraining Stage

The goal of pretraining is to enable the model to extract feature representations from a mass of unlabeled data, and

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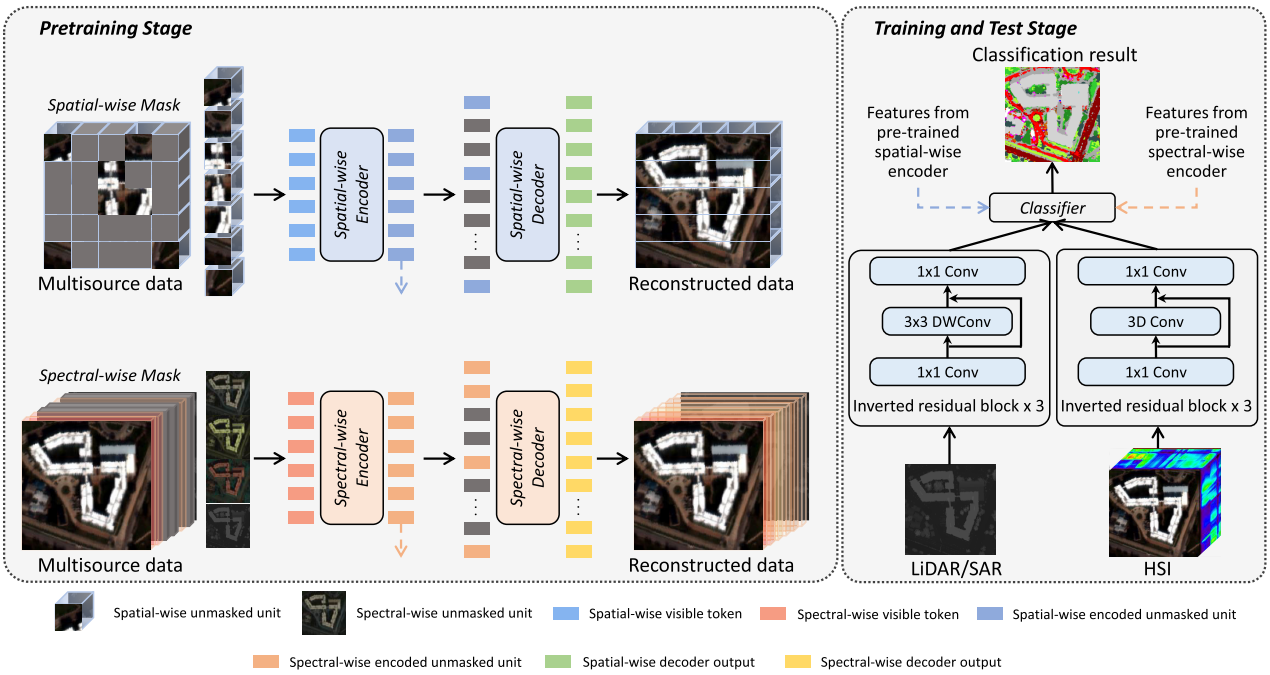


Fig. 2. Framework of the proposed SS-MAE for multisource image classification. It is comprised of the pretraining stage, the training stage, and the testing stage. In the pretraining stage, the network contains a spatialwise branch and a spectralwise branch. The spatialwise branch masks random patches and reconstructs missing pixels, while the spectralwise branch masks random spectral channels and reconstructs missing channels. In the training and test stages, two lightweight CNNs are built to complement local features. Each lightweight CNN contains three IRBs, and therefore, both global and local features are used in feature representation.

MAE confirms that mask reconstruction can easily achieve this. In this article, we pretrain the model using the similar mask reconstruction as MAE. Algorithm 1 provides the pseudocode of SS-MAE in pretraining stage.

Algorithm 1 Pseudocode of SS-MAE in a PyTorch-Like Style (Pretraining Stage)

```
# Spa_E, Spe_E: Spatial-wise and Spectral-wise encoders
# Spa_D, Spe_D: Spatial-wise and Spectral-wise decoders
# x1: SAR or LiDAR
# x2: HSI

# load a mini-batch x1 and x2 with N samples
# x1: N x c1 x P x P
# x2: N x c2 x P x P
for x1, x2 in loader:
# T: N x (c1 + c2) x P x P
T = concat([x1, x2], dim = 1)
TM_spa = spa_mask(T) # spacial-wise mask
TM_spe = spe_mask(T) # spectral-wise mask

#f_spa: N x (P x P) x D, f_spe: N x (c1 + c2) x D
f_spa = Spa_E.forward(TM_spa.view(N, P * P, c1 + c2))
f_spe = Spe_E.forward(TM_spe.view(N, c1 + c2, P * P))

# tr: linear projection and resize
# R_spa: N x (c1 + c2) x P x P
R_spa = tr(Spa_D.forward(f_spa))
# R_spe: N x (c1 + c2) x P x P
R_spe = tr(Spe_D.forward(f_spe))

# MSE loss computation
loss = MSELoss(R_spa, T) + MSELoss(R_spe, T)

# back propagation
loss.backward()
update(Spa_E.param, Spa_D.param, \
Spe_E.param, Spe_D.param)
```

1) Masking Strategy: MIM is one of the most prominent pretext tasks in self-supervised learning methods. During pretraining, MIM is implemented by masking a portion of the image and asking the model to reconstruct the image representation or feature representation being masked, thus forcing the model to learn universal features in unlabeled data. Random masking is an essential component of the MIM technique. However, the existing methods are often limited to masking along the spatial dimension, ignoring the spectral information. To this end, we propose spatial–spectral masking method for hyperspectral feature representation.

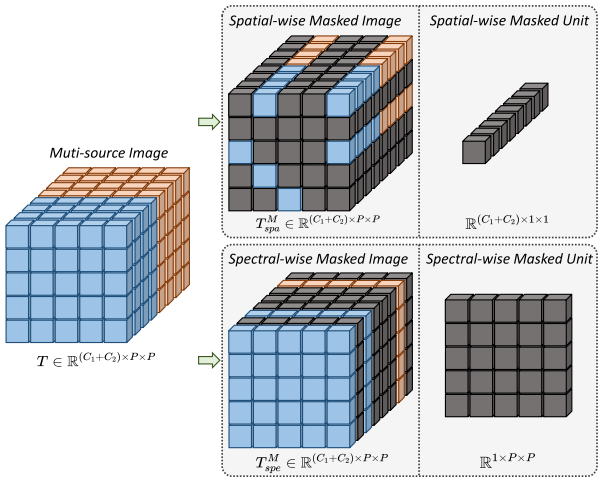


Fig. 3. Masking strategy of spatialwise masking and spectralwise masking.

Spatialwise Masking: We randomly mask some pixels in the spatial dimension, as shown in Fig. 3. For the input $T \in \mathbb{R}^{(C_1+C_2)\times P\times P}$, the size of each masked unit is $(C_1+C_2)\times 1\times 1$. Hence, we obtain the spatialwise masked image T^M_{spa}.

Spectralwise Masking: As the responses of different spectral bands for the same object are sometimes similar, the challenge

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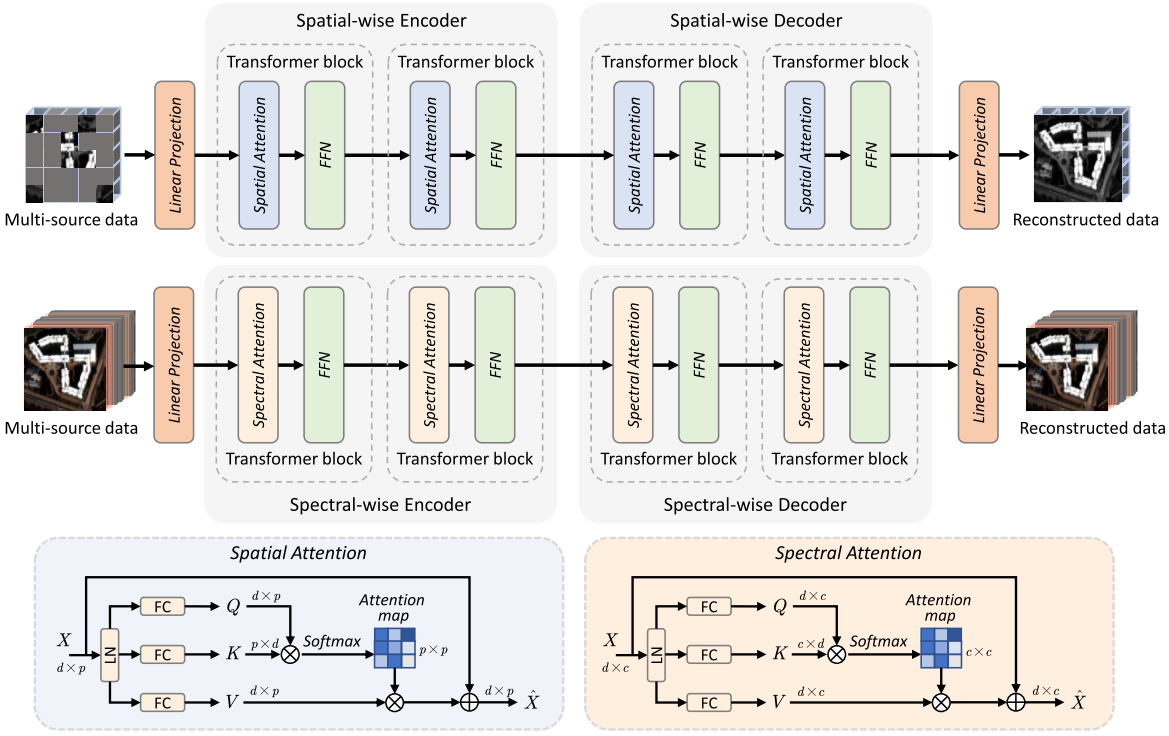


Fig. 4. Architecture of the dual-attention Transformer. Spatialwise attention and spectralwise attention are considered for masked information reconstruction. Therefore, the model fully exploits the spatial and spectral representations of the input multisource data.

of reconstruction is reduced if different spectra are spatially complementary to each other. As such, we mask the whole band data for spectralwise masking. Specifically, for the input $T \in \mathbb{R}^{(C_1+C_2)\times P\times P}$, we compute a spectralwise masked image T^M_{spe} by randomly masking some spectral bands. The size of each masked unit is $1\times P\times P$, as shown in Fig. 3. The masking ratio is a critical parameter that may affect the classification performance. Therefore, we conducted comprehensive experiments in Section IV-B to select the best masking ratio.

It should be noted that spectralwise masking is not applicable to single-band data, such as LiDAR. Consequently, for HSI and LiDAR data classification, we deliberately exclude LiDAR in spectralwise masking. However, for HSI and multispectral data joint classification, or HSI and PolSAR data joint classification, spectralwise masking is essential for the proposed SS-MAE.

2) Reconstruction Target: Our SS-MAE reconstructs the input by predicting the pixel values in spatialwise and spectralwise masked units. The loss function consists of a spatial reconstruction loss $\mathcal{L}(T, R_{\text{spa}})$ and a spectral reconstruction loss $\mathcal{L}(T, R_{\text{spe}})$. The whole loss $\mathcal{L}_{\text{SSMAE}}$ is a weighted sum

$$\mathcal{L}_{\text{SSMAE}} = \lambda\mathcal{L}(T, R_{\text{spa}}) + \mathcal{L}(T, R_{\text{spe}}). \tag{1}$$

We use mean-squared error (MSE) loss for $\mathcal{L}(T, R_{\text{spa}})$ and $\mathcal{L}(T, R_{\text{spa}})$. λ is set to 2 in our experiments. In order to maintain consistency with the loss function of MAE, we only compute the loss pertaining to masked pixels, while disregarding those that are unmasked.

C. Dual-Attention Transformer

Leveraging the natural property that both masked and unmasked units can serve as input tokens, we propose a dual-attention Transformer encoder–decoder architecture to accomplish the task of reconstructing T^M_{spa} and T^M_{spe} into T, as shown in Fig. 4.

1) Spatialwise Reconstruction: Only visible tokens in T^M_{spa} are retained, and these tokens are handled by linear projection to generate $T^{M'}_{\text{spa}} \in \mathbb{R}^{d\times p}$. Here, p is the number of visible tokens, and d is the feature dimension of each token. Next, $T^{M'}_{\text{spa}}$ is fed into the spatialwise encoder with $\mathcal{B}$ Transformer blocks to compute the encoded visible tokens.

After that, full image tokens (consisting of both encoded visible tokens and mask tokens) are fed into the spatialwise decoder to accomplish the reconstruction task. Positional embeddings are used to encode the token locations in the image. The decoder has another $\mathcal{B}$ Transformer blocks.

Each Transformer block is composed of spatialwise attention computation and feedforward neural network (FFN). For spatialwise attention, given $X \in \mathbb{R}^{d\times p}$ from the input, we first apply 1×1 convolutions to enhance the input feature and obtain query Q, key K, and value V. After that, Q and K are reshaped to conduct dot product to generate spatialwise attention map $A \in \mathbb{R}^{p\times p}$. Overall, the spatial attention is defined as follows:

$$\hat{X} = \text{Attention}\big(\hat{Q}, \hat{K}, \hat{V}\big) + X \tag{2}$$

$$\text{Attention}\big(\hat{Q}, \hat{K}, \hat{V}\big) = \hat{V}\,\text{Softmax}\big(\hat{K}\hat{Q}/\alpha\big) \tag{3}$$

where X and $\hat{X}$ are the input and output feature maps. $\hat{Q}$, $\hat{K}$, and $\hat{V}$ are reshaped tensors from the input. α is a learnable scaling parameter to control the softmax. To achieve multihead attention, we split the Q, K, and V into $\mathcal{H}$ heads along the feature channel dimension, respectively. The dimension of each head is $p/\mathcal{H}$, and hence, separate attention maps can be learned efficiently in parallel. We adopt the identical FFN

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architecture as proposed in ViT [41]. This FFN consists of two linear layers, with a Gaussian error linear unit (GELU) activation function separating them. The first layer expands the dimension by a factor of 4, while the subsequent layer reduces it by the same ratio.

2) Spectralwise Reconstruction: Here, visible tokens are spectral bands, and the mask tokens are masked spectral bands. Similarly, only visible tokens are retained and fed into the encoder to compute the encoded visible tokens. In spectralwise decoder, both the visible tokens and mask tokens are fed into the decoder for the reconstruction task.

It should be noted that after linear projection, T_{spa}^{M} is transformed to $T_{\text{spe}}^{M'} \in \mathbb{R}^{d\times c}$. Here, c is the number of visible spectral bands, and d is the feature dimension of each band after linear projection. In spatial attention, the attention map computes the similarity of visible pixels, while in spectral attention, the attention map computes the similarity of visible spectral bands.

D. Training and Optimization

Conventional masked autoencoders only fine-tune the encoder parameters on the training data. Although it is convenient and easy to implement, the model may perform suboptimally for hyperspectral data. Specifically, Transformer tends to pay more attention on long-range dependencies of the features and the semantics of the input data. Some local features or high-frequency information (such as edge and texture) may get lost.

In order to add local features, we add two lightweight CNN to capture local features. For LiDAR/SAR data, the network consists of three IRBs. As shown in Fig. 2, the IRB is composed of a projection layer, a depthwise convolution layer, and an expansion projection layer. It is defined as follows:

$$\text{IRB}(X) = W_1(\mathcal{F}(W_2(X))) \tag{4}$$

$$\mathcal{F}(X) = \text{DWConv}(X) + X \tag{5}$$

where W_1 and W_2 are 1×1 convolution and DWConv$(\cdot)$ is 3×3 depthwise convolution. We include the GELU activation and batch normalization after the first and second convolution layers. The depthwise convolution is capable of extracting local features with negligible extra computational cost.

For HSI feature extraction, the IRB is composed of a projection layer, a 3-D convolution layer, and an expansion projection layer. The difference from IRB for LiDAR/SAR data is the replacement of the depthwise convolution with a 3-D convolution. The expansion ratio is 2 in our implementations.

Finally, the features from spatialwise and spectralwise Transformer encoders are stacked with features from IRBs, as shown in Fig. 2. Both global and local information from the input multisource data are integrated for classification. Algorithm 2 shows the pseudocode of SS-MAE for the training stage.

Algorithm 2 Pseudocode of SS-MAE in a PyTorch-Like Style (Training Stage)

```
# Spa_E, Spe_E: Spatial-wise and Spectral-wise encoders
# x1: SAR or LiDAR
# x2: HSI
# labels: labels in the dataset
# IRB: Inverted residual block

# load a mini-batch x1 and x2 with N samples
# x1: N x c1 x P x P
# x2: N x c2 x P x P
for x1, x2 in loader:
# T: N x (c1 + c2) x P x P
T = concat([x1, x2], dim = 1)

# f_spa: N x (PxP) x D, f_spe: N x (c1 + c2) x D
f_spa = Spa_E.forward(T.view(N, P * P, c1 + c2))
f_spe = Spe_E.forward(T.view(N, c1 + c2, P * P))

# tr: linear projection and resize
I_1 = tr(IRB_1.forward(x1)) # I_1: N x 256~x D
I_2 = tr(IRB_2.forward(x2)) # I_2: N x 256~x D

# fn: N x (c1 + c2 + PxP + 512) x D
fn = concat(f_spa + f_spe + I_1 + I_2)

# r: N x Cnum, Cnum is the number of categories
r = classifier(fn)

# CrossEntropy loss computation
loss = CrossEntropyLoss(r, labels) # labels: N x Cnum

# back propagation
loss.backward()
update(Spa_E.param, Spe_E.param, \
IRB_1.param, IRB_2.param, classifier.param)
```

IV. Experimental Results and Analysis

A. Dataset Description

To evaluate the effectiveness of our proposed SS-MAE, we employ it on three multisource remote sensing datasets: Berlin, Augsburg, and Houston 2018. Berlin and Augsburg datasets are used for hyperspectral and SAR data classification. Houston 2018 is used for hyperspectral and LiDAR data classification.

1) Berlin Dataset: It covers the urban and rural areas of Berlin. The hyperspectral data are simulated environmental mapping and analysis program (EnMAP) data based on HyMap HS data. The SAR data are Sentinel-1 dual-Pol single-look complex (SLC) product obtained from the European Space Agency (ESA) [51]. The HSI has 797 $\times$ 220 pixels and 244 spectral bands in the wavelength range of 400–2500 nm [52]. The SAR image has 1723 $\times$ 476 pixels. The nearest neighbor interpolation was employed to match the spatial resolution of both images.

2) Augsburg Dataset: This dataset is composed of HSI and SAR data from Augsburg, Germany. The HSI was acquired by the HySpex sensor, and the SAR data were acquired by the Sentinel-1 sensor. All images are with the ground sample distance (GSD) of 30 m. The spatial size of both images is 332 $\times$ 485 pixels. The HSI has 80 spectral bands ranging from 0.4 to 2.5 μm, and the SAR data have four features derived from polarization decomposition (VV intensity, VH intensity, real part, and imaginary part of the off-diagonal element of the PolSAR covariance matrix).

3) Houston 2018 Dataset: This dataset covers the University of Houston campus and the neighboring urban area. The HSI contains 48 bands in the wavelength range of 380–1050 nm. LiDAR data are a multispectral LiDAR image with three bands. The University of Houston released the dataset as part of the 2018 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. We use the training subset of the whole dataset in this article.

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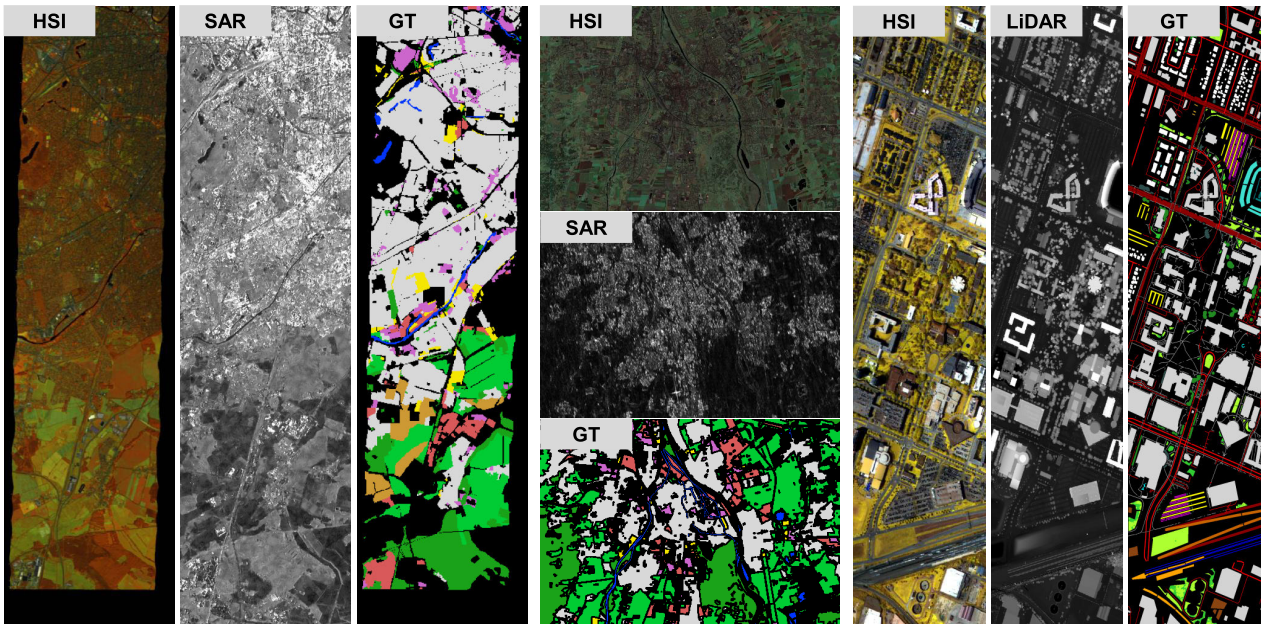


Fig. 5. Visualization of the datasets. (a) Berlin dataset. (b) Augsburg dataset. (c) Houston 2018 dataset.

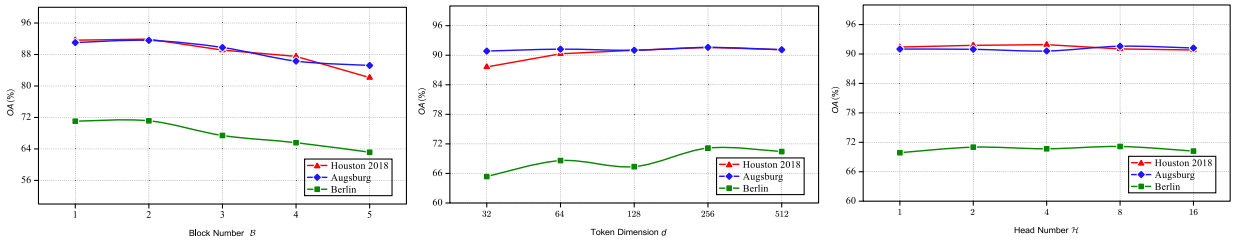


Fig. 6. OA (%) achieved by the proposed SS-MAE with different parameters for the Berlin, Augsburg, and Houston 2018 datasets. (a) Block number versus OA. (b) Token dimension versus OA. (c) Head number versus OA.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE BERLIN DATASET

No.	Name	Color	Train	Test
1	Forest		443	54511
2	Residential area		423	268219
3	Industrial area		499	19067
4	Low plants		376	58906
5	Soil		331	17095
6	Allotment		280	13025
7	Commercial area		298	24526
8	Water		170	6502
	Total		2820	461851

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE AUGSBURG DATASET

No.	Name	Color	Train	Test
1	Forest		146	13361
2	Residential area		264	30065
3	Industrial area		21	3830
4	Low plants		248	26609
5	Allotment		52	523
6	Commercial area		7	1638
7	Water		23	1507
	Total		761	77533

Tables I–III enumerate the number of samples for both training and testing on three datasets. Fig. 5 displays the HSI data in false color, along with the SAR/LiDAR images and the ground truth. The evaluation metrics used in this article include overall accuracy (OA), average accuracy (AA), and Kappa. Specifically, OA denotes the proportion of accurately classified samples throughout the entirety of the dataset, AA signifies the mean accuracy across all classes present within the dataset, and Kappa serves as a statistical measure of the concurrence between predicted and true labels.

B. Parameter Settings

This article investigates the influence of parameter settings on classification performance for remote sensing image classification tasks and mask-reconstruction tasks. We fine-tune two categories of parameters: network structure (the number of blocks and heads in the Transformer encoder–decoder, and the size of feature dimension) and input data (input patch size, the number of principal components, the pretraining sample size, the spatial masking ratio, and the channel masking ratio).

Parameters of the Network Structure: To expedite experimentation, we maintain identical network structure parameters for the Transformer decoder and encoder throughout pretraining and avoid pretrained weights in preference to training from scratch in the experimental process for assessing the influence of network structure parameters.

1) Block Number: Fig. 6(a) shows the effect of different block number $\mathcal{B}$ values on the classification performance of three datasets. When $\mathcal{B} = 2$, our model achieved the

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TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON 2018 DATASET

No.	Name	Color	Train	Test
1	Health grass		1000	39196
2	Stressed grass		1000	130008
3	Artificial turf		1000	2736
4	Evergreen trees		1000	54322
5	Deciduous trees		1000	20172
6	Bare earth		1000	18064
7	Water		500	1064
8	Residential buildings		1000	15899
9	Non-residential buildings		1000	894769
10	Roads		1000	183283
11	Sidewalks		1000	136035
12	Crosswalks		1000	6059
13	Major thoroughfares		1000	185438
14	Highways		1000	39438
15	Railways		1000	27748
16	Paved parking lots		1000	45932
17	Unpaved parking lots		250	587
18	Cars		1000	26289
19	Trains		1000	21479
20	Stadium seats		1000	27296
	Total		18750	2018910

highest OA values, which were 71.15%, 91.59%, and 91.91%, respectively. It indicates that two blocks are sufficient to capture the essential features from the input data and enhance the representation power of SS-MAE.

2) Token Dimension: We test the optimal token dimension d for each dataset. Fig. 6(b) reveals the relationship between different d values and the classification performance of the proposed SS-MAE for each dataset. It shows that when $d = 256$, our model achieved the best OA values on all three datasets. It demonstrates that 256-D feature for one token is optimal for the semantic information extraction from the input data.

3) Heads Number: We test the performance of the proposed SS-MAE with the head number $\mathcal{H}$ from 1 to 16. Fig. 6(c) shows how different $\mathcal{H}$ values affect the classification performance on three datasets. The best $\mathcal{H}$ values for Berlin, Augsburg, and Houston 2018 datasets are 8, 8, and 4, respectively. Increasing the number of heads $\mathcal{H}$ does not significantly enhance performance and sometimes even degrades the classification accuracy.

Parameters of the Input Data: In addition to the regular parameters of the input patch size and the number of principal components, we pay special attention to three parameters against training data in pretraining stage: the pretraining sample size, the spatial masking ratio, and the channel masking ratio. We keep the spatial masking ratio and channel masking ratio consistent during pretraining.

TABLE IV
PERFORMANCE OF THE PROPOSED MODEL WITH DIFFERENT PRETRAINING SAMPLE NUMBERS

Pre-training sample number	Overall Accuracy (%)		
	Berlin	Augsburg	Houston 2018
1000	71.32	92.11	91.93
10000	71.59	92.21	91.89
50000	71.91	**92.27**	**92.43**
100000	72.36	92.26	92.25
200000	**73.43**	-	92.26

TABLE V
PERFORMANCE OF THE PROPOSED MODEL WITH DIFFERENT MASKING RATIOS

Masking Ratio	Overall Accuracy(%)		
	Berlin	Augsburg	Houston 2018
0.1	72.08	92.17	91.99
0.3	**73.43**	92.24	92.32
0.5	71.85	92.25	**92.43**
0.7	72.08	**92.27**	92.29
0.9	72.36	92.20	92.15

1) Input Patch Size: We test five input patch sizes: 5×5, 7×7, 9×9, 11×11, and 13×13. The input patch size determines the spatial receptive fields captured by each token. As shown in Fig. 7(a), the optimal input patch size for all three datasets is 7×7. This indicates that a moderate amount of spatial information is sufficient for our model to learn effective representations, and larger patches increase the redundancy of information.

2) Number of Principal Components: We use principal component analysis (PCA) to reduce the spectral redundancy of the input HSI. We test different numbers of principal components $\mathcal{K}$ from 15 to 35. Fig. 7(b) shows how $\mathcal{K}$ affects the classification performance on different datasets. It indicates that $\mathcal{K} = 30$ is optimal for Berlin Dataset and Augsburg Dataset, while $\mathcal{K} = 20$ is optimal for Houston 2018 Dataset. Therefore, we draw the conclusion that different datasets have different levels of spectral complexity and redundancy.

3) Pretraining Sample Number: For pretraining, we randomly sample a subset of pixels from each dataset. Table IV illustrates how increasing the sample number $\mathcal{A}$ improves the classification performance of our model on each dataset. We found that $\mathcal{A} = 50\,000$ is optimal for Augsburg Dataset and Houston 2018 Dataset, while $\mathcal{A} = 200\,000$ is optimal for Berlin Dataset. Overall, we observe that a larger sample set leads to better performance. Note that the classification accuracy tends to decrease slightly when $\mathcal{A}$ is greater than 50 000 on Augsburg and Houston 2018 datasets. The optimal sample number for pretraining depends on the diversity of each dataset.

4) Spatial/Channel Masking Ratio: We use two types of masking strategies during pretraining: spatial masking (corrupting contiguous regions) and channel masking (corrupting entire spectral bands). We keep both masking ratios consistent during pretraining. We test different masking ratios from 0.1 to 0.9. Table V depicts the relationship between masking ratio and classification accuracy on three datasets. The Berlin, Augsburg, and Houston 2018 datasets require

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TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE BERLIN DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	Fusion-HCT	SS-MAE*	SS-MAE
Forest	81.75	86.24	81.09	80.90	77.40	78.01	74.58	79.53	79.35
Residential area	76.26	91.38	73.05	72.81	71.30	74.05	72.58	70.99	76.04
Industrial area	39.67	19.76	62.61	38.89	38.98	39.48	48.92	63.06	68.51
Low plants	49.78	20.00	82.82	78.09	79.30	84.15	83.04	80.19	79.17
Soil	89.42	48.72	86.41	73.48	87.69	88.03	75.24	92.87	91.59
Allotment	54.36	38.89	54.61	72.54	81.65	70.00	76.34	64.60	56.86
Commercial area	4.65	18.47	2.56	22.80	36.52	38.18	29.66	26.37	19.75
Water	41.93	29.61	75.96	68.15	75.96	56.41	78.50	68.40	66.87
OA	67.60	70.91	71.08	70.33	70.82	72.63	71.18	71.15	73.43
AA	54.72	44.13	64.88	63.45	68.59	66.04	67.36	68.24	67.26
Kappa	50.96	51.07	58.02	56.90	58.53	60.48	58.66	59.00	60.91

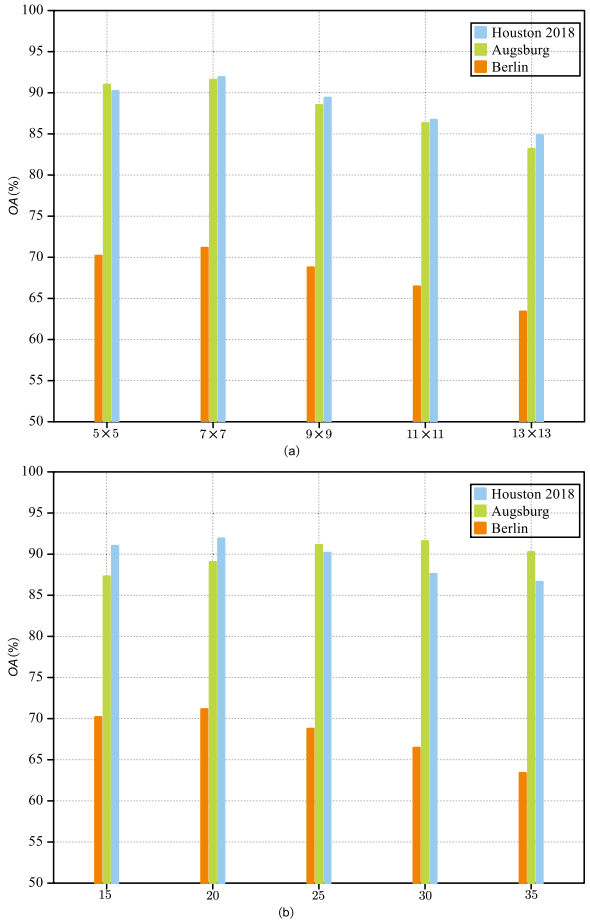


Fig. 7. OA (%) achieved by the proposed SS-MAE with different parameters for the Berlin, Augsburg, and Houston 2018 datasets. (a) Input patch size $P \times P$. (b) Number of principal component $\mathcal{K}$.

the optimal masking ratios of 0.3, 0.7, and 0.5, respectively. It is evident that with a higher masking ratio, the model learns more robust feature representations via sparse data reconstruction. Nevertheless, if the ratio is too high, the model may fail to restore comprehensive information and impair the classification performance.

C. Performance Comparison

We compare the proposed SS-MAE with the existing state-of-the-art methods, including two branch CNN (TBCNN) [23], FusAtNet [53], spatial-spectral enhancement network (S^2ENet) [54], depthwise feature interaction network (DFINet) [32], AsyFFNet [55], ExViT [56], and fusion with hierarchical CNN and Transformer (Fusion-HCT) [57]. TBCNN investigates the classification fusion of hyperspectral imagery and data from other sensors using a two-branch convolution neural network, which extracts spectral–spatial features from HSI and other data features from LiDAR or high-resolution visual images. FusAtNet proposes a multisource classification framework by utilizing a self-attention mechanism for spectral features and a cross-attention approach for spatial features. S^2ENet proposes a spatial–spectral enhancement module for cross-modal information interaction, which effectively facilitates the interaction and understanding between hyperspectral and LiDAR data. DFINet uses a depthwise cross-attention module to extract complementary information from multisource feature pairs. AsyFFNet is a multisource data classification method based on asymmetric feature fusion, which employs weight-share residual blocks for feature extraction and a feature calibration module for the spatialwise multisource feature modeling. ExViT utilizes parallel branches of position-shared Transformer extended with separable convolution modules to process multimodal image patches. A cross-modality attention module is employed to fuse tokenized embeddings. Fusion-HCT combines hierarchical CNN and Transformer network to extract multisource data features and then fuses both sets of features by cross-token attention fusion encoder.

1) Results on the Berlin Dataset: The classification performance of different methods on the Berlin dataset is shown in Table VI. SS-MAE* is the proposed model without pretraining. Compared with other methods, SS-MAE* and SS-MAE achieve better classification performance in Industrial area class and Soil class. The OA improvements achieved by SS-MAE* are 3.55%, 8.7%, 0.24%, 0.07%, 0.82%, and 0.33% compared with TBCNN, FusAtNet, S^2ENet, DFINet, and AsyFFNet [55], respectively. It can be observed from Table VI that SS-MAE* exhibits a decrease of 1.48% compared with ExViT and a marginal reduction of 0.03% compared with Fusion-HCT. With the help of pretraining, SS-MAE not only surpasses ExViT and Fusion-HCT, but also attains the best classification performance. The improvement in performance

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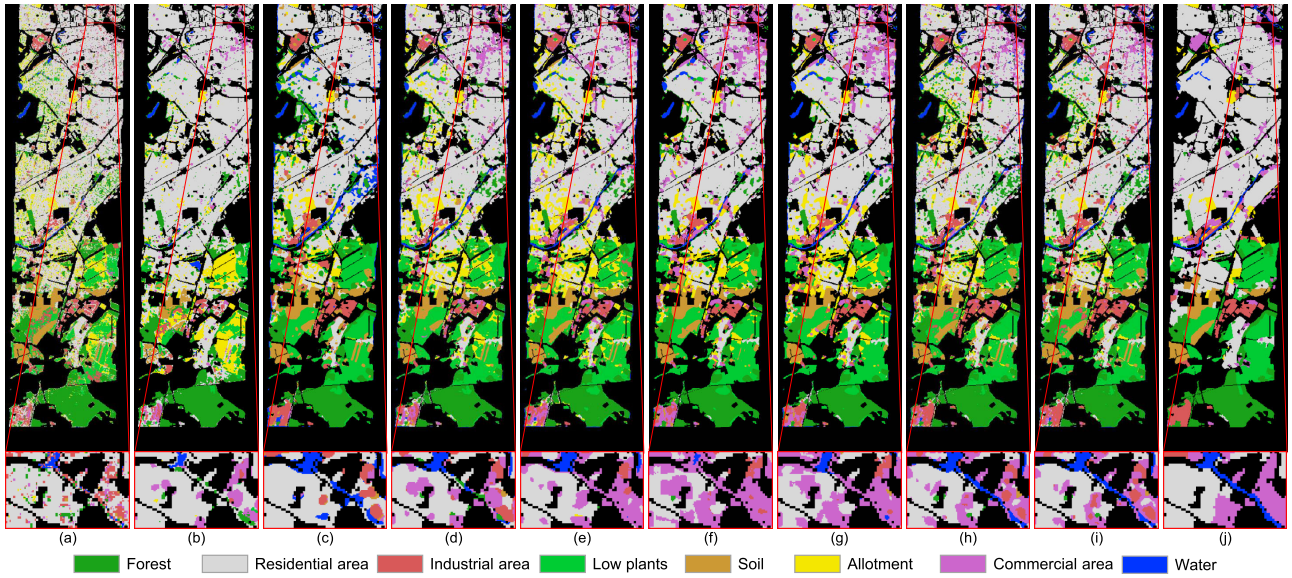


Fig. 8. Classification results of different methods for the Berlin dataset. (a) TBCNN. (b) FusAtNet. (c) S^2ENet. (d) DFINet. (e) AsyFFNet. (f) ExViT. (g) Fusion-HCT. (h) SS-MAE. (i) SS-MAE (pretrained). (j) Ground truth.

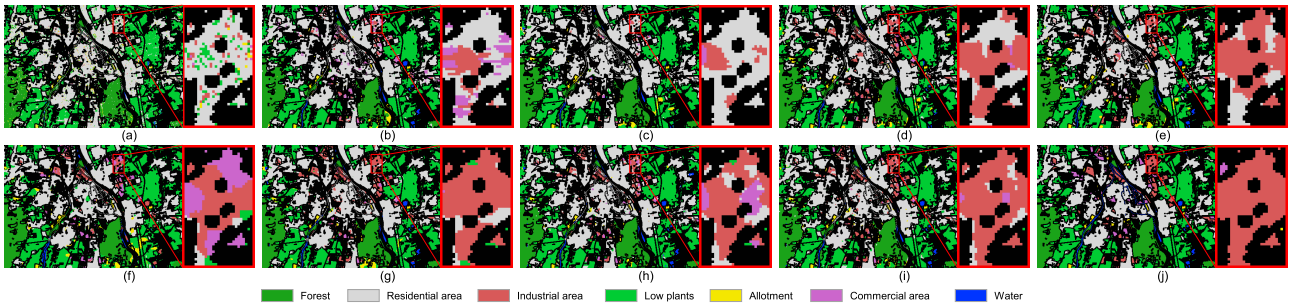


Fig. 9. Classification results of different methods for the Augsburg dataset. (a) TBCNN. (b) FusAtNet. (c) S^2ENet. (d) DFINet. (e) AsyFFNet. (f) ExViT. (g) Fusion-HCT. (h) SS-MAE. (i) SS-MAE (pretrained). (j) Ground truth.

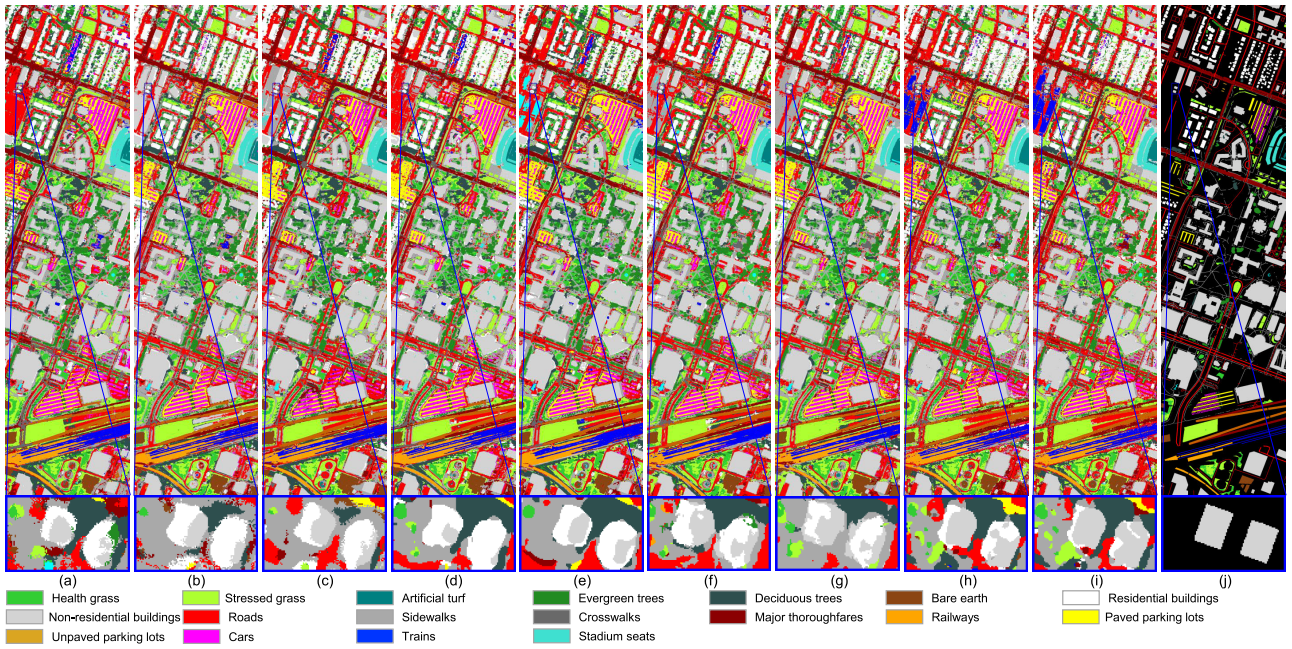


Fig. 10. Classification results of different methods for the Houston 2018 dataset. (a) TBCNN. (b) FusAtNet. (c) S^2ENet. (d) DFINet. (e) AsyFFNet. (f) ExViT. (g) Fusion-HCT. (h) SS-MAE. (i) SS-MAE (pretrained). (j) Ground truth.

can be attributed to the model's comprehensive integration of contextual and semantic information during the pretraining phase. With pretraining, SS-MAE is able to capture more universal features from extensive quantities of unlabeled data. These universal features enable the proposed SS-MAE to develop a more profound comprehension of the input multisource data, thereby resulting in an enhancement of classification accuracy. The visualized classification results are

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TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	Fusion-HCT	SS-MAE*	SS-MAE
Forest	90.88	93.78	98.10	97.38	96.01	90.04	97.32	96.82	96.48
Residential area	93.89	97.58	99.08	98.37	97.96	95.44	98.28	98.14	97.77
Industrial area	8.28	26.48	12.19	61.31	53.29	34.58	67.12	55.32	74.77
Low plants	91.97	97.67	91.78	92.63	91.74	90.68	93.33	95.94	95.87
Allotment	38.24	52.77	45.12	49.33	65.39	51.82	56.41	45.80	48.09
Commercial area	1.40	24.66	1.22	3.54	4.58	28.63	8.42	14.29	9.45
Water	10.82	47.51	24.09	26.61	49.64	17.65	49.77	46.21	43.52
OA	84.53	90.62	88.22	90.66	90.15	86.65	91.75	91.59	92.27
AA	47.92	62.92	53.08	61.30	65.51	58.40	67.24	64.64	66.56
Kappa	77.13	86.33	82.61	86.47	85.74	80.79	88.12	87.67	88.68

TABLE VIII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON 2018 DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	Fusion-HCT	SS-MAE*	SS-MAE
Health grass	94.84	96.28	90.93	94.29	96.90	93.65	98.44	90.53	94.00
Stressed grass	92.60	93.45	94.76	92.71	91.82	95.44	91.05	95.56	95.34
Artificial turf	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Evergreen trees	98.80	98.33	98.52	98.92	99.29	98.52	97.39	98.23	98.55
Deciduous trees	97.12	99.11	99.39	97.77	99.52	99.25	98.41	96.62	98.15
Bare earth	99.61	100.00	99.96	100.00	99.98	99.92	100.00	100.00	99.84
Water	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Residential buildings	93.21	97.90	97.16	97.36	95.16	96.88	96.80	95.89	94.98
Non-residential buildings	91.30	93.41	94.93	93.29	95.02	94.87	94.74	95.85	96.27
Roads	61.06	74.56	74.58	75.73	72.72	80.51	76.99	78.77	82.97
Sidewalks	75.91	82.94	79.47	83.67	77.18	80.27	83.14	78.29	74.10
Crosswalks	85.31	84.65	96.36	94.23	93.14	96.76	93.89	97.78	89.57
Major thoroughfares	72.77	86.90	83.50	81.20	84.67	82.11	86.29	80.20	83.21
Highways	95.86	97.59	98.57	98.91	99.53	84.10	93.75	98.28	98.08
Railways	99.78	99.71	99.84	99.94	99.90	99.81	99.86	100.00	99.73
Paved parking lots	90.74	98.08	97.41	98.37	99.18	99.08	98.62	95.36	96.36
Unpaved parking lots	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Cars	98.47	97.11	99.17	99.09	96.64	97.56	95.78	98.19	97.20
Trains	99.90	99.63	100.00	99.46	100.00	99.90	100.00	99.46	100.00
Stadium seats	99.92	99.93	99.98	99.98	99.89	99.94	99.97	100.00	100.00
OA	86.95	91.52	91.61	91.02	91.24	91.87	91.98	91.91	92.43
AA	92.36	94.98	95.22	95.24	95.02	94.93	95.26	94.94	94.91
Kappa	83.39	89.14	89.17	88.46	88.71	89.51	89.66	89.51	90.19

shown in Fig. 8. It is evident that the proposed SS-MAE effectively distinguishes the Industrial area and Residential area, as well as Allotment and Low plants.

2) Results on the Augsberg Dataset: Table VII illustrates the classification performance of different methods on the Augsburg dataset. In comparison with other methods, SS-MAE* achieves the best OA and Kappa values except for slightly lower than Fusion-HCT. Furthermore, the classification performance of SS-MAE is further enhanced via pretraining, resulting in an increase of 0.68% in OA value. The visualized classification results are shown in Fig. 9. It demonstrates that the proposed SS-MAE can correctly classify the Industrial area and Residential area, which is a difficult task for the other methods.

3) Results on the Houston 2018 Dataset: Table VIII illustrates the classification performance of various models on the Houston 2018 dataset. Without pretraining, SS-MAE* achieves the second-best OA value. Pretraining the SS-MAE model

TABLE IX
QUANTITATIVE COMPARISON OF MODEL COMPLEXITY IN TERMS OF THE NUMBER OF PARAMETERS AND GFLOPS

Model	Params. (MB)	GFLOPs
TBCNN	0.1059	0.6567
FusAtNet	36.1518	305.9769
S^2ENet	0.1629	1.7582
DFINet	0.7968	5.4959
AsyFFNet	1.0632	8.6556
ExViT	0.2270	2.9085
Fusion-HCT	0.4310	0.8057
SS-MAE	4.4000	15.0728

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TABLE X
INFLUENCE OF SPECTRAL ATTENTION, SPATIAL ATTENTION, AND IRB ON CLASSIFICATION RESULTS OF SS-MAE

Spectral attention	Spatial attention	IRB	Berlin	Augsburg	Houston2018
✔	✖	✖	67.42	86.04	86.73
✖	✔	✖	70.73	89.92	90.81
✔	✔	✖	72.35	90.91	91.73
✔	✔	✔	73.43	92.27	92.43

TABLE XI
FEW-SHOT CLASSIFICATION EXPERIMENTS OF SS-MAE* AND SS-MAE ON HOUSTON 2018 DATASET

Number of samples	SS-MAE*	SS-MAE
1	39.01	40.22
2	39.91	42.37
4	54.17	55.57
8	61.39	62.25
16	67.92	68.97

further improves its performance, with an increase of 0.52% in OA. The classification maps of different methods are shown in Fig. 10. Major Thoroughfares and Roads are challenging classes, since their spectral responses are similar. It is evident that the proposed SS-MAE shows fewer misclassifications between both classes.

4) Analysis of Computational Complexity: Table IX shows the computational complexity of different methods on the Houston 2018 dataset. It can be observed that the proposed method ranks second in terms of both parameters (Params) and giga floating-point operations (GFLOPs). It means that the computational complexity of the proposed SS-MAE surpasses most of the existing deep learning-based models. The reason is that SS-MAE is a typical unsupervised method, harnessing rich spectral and spatial representations via unsupervised learning. The larger model generalizes better for unsupervised learning. Therefore, the proposed SS-MAE uses a Transformer-like architecture with 4.4-MB parameters to obtain enhanced spectral and spatial representations. In practical scenarios, our model can concurrently compute multihead attentions, thereby enhancing both training efficiency and inference speed.

D. Ablation Study

1) Effectiveness of Essential Components: To evaluate the effectiveness of the essential components of our proposed SS-MAE, we conduct ablation studies on three datasets. The ablation studies aim to investigate the impact of spatial attention, spectral attention, and IRB. Table X exhibits the relationship between different components and OA values. When employing only spatial attention, the OA values on the Berlin, Augsburg, and Houston 2018 datasets are 67.42% 86.04%, and 86.73%, respectively. However, when solely using spectral attention, the OA values on three datasets are 3.31%, 3.88%, and 4.08% higher. It indicates that for HSI interpretation, the spectral information is critical and is an important factor for land cover classification. When simultaneously using spatial and spectral attention, the classification performance is further improved. It reflects the essential role of the two-branch structure of the proposed SS-MAE, and SS-MAE can simultaneously capture the rich spatial and spectral features. Furthermore, for the sake of balancing global and local features, we add IRB for local feature extraction. The IRB increases the classification performance by 1.08%, 1.36%, and 0.7% on the three datasets, respectively. It demonstrates the effectiveness of supplementing local features with lightweight CNNs.

2) Few-Shot Classification: To verify the effectiveness of pretraining, we compare the classification performance of SS-MAE* with SS-MAE using small number of training samples. Specifically, we train two models on the Houston2018 dataset, using randomly picked 2, 4, 8, and 16 samples, respectively. The results presented in Table XI show that pretraining consistently boosts the classification performance of the proposed SS-MAE.

V. CONCLUSION

In this article, we propose a novel framework, SS-MAE, which is an SS-MAE for multisource remote sensing image classification. In particular, it consists of spatialwise branch and spectralwise branch. The spatial branch masks random patches and reconstructs missing pixels, while the spectralwise branch masks random spectral channels and reconstructs missing channels. Therefore, the proposed SS-MAE exploits both the spatial and spectral feature representations of the input hyperspectral data. Furthermore, to complement local features in the training stage, we add two lightweight CNNs for feature extraction. Both global and local features are used in feature modeling. Extensive experiments on the dataset demonstrate the superiority of the proposed SS-MAE over several state-of-the-art methods. In particular, we find that the proposed SS-MAE has excellent performance even without pretraining. The Transformer and CNN hybrid architecture benefit the cross-modal feature representation.

In the future, we plan to explore a more diverse cross-modal feature fusion strategy. As a pioneer study, we also plan to conduct more experiments on MIM for multisource remote sensing data classification.

REFERENCES

[1] H. Li, B. Zhou, and F. Xu, "Variation analysis of spectral characteristics of reclamation vegetation in a rare Earth mining area under environmental stress," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408412.

[2] B. Cui, X. Li, J. Wu, G. Ren, and Y. Lu, "Tiny-scene embedding network for coastal wetland mapping using Zhuhai-1 hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

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[3] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence-induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510213.

[4] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[5] S. Wang, D. Hou, and H. Xing, "A self-supervised-driven open-set unsupervised domain adaptation method for optical remote sensing image scene classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605515.

[6] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[7] C. Wang, L. Zhang, W. Wei, and Y. Zhang, "Dynamic super-pixel normalization for robust hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505713.

[8] F. Luo, L. Zhang, B. Du, and L. Zhang, "Dimensionality reduction with enhanced hybrid-graph discriminant learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5336–5353, Aug. 2020.

[9] J. Shi and H. Jin, "Riemannian nearest-regularized subspace classification for polarimetric SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] J. Cheng, D. Xiang, Q. Yin, and F. Zhang, "A novel crop classification method based on the tensor-GCN for time-series PolSAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5237614.

[11] J. Ni, C. López-Martínez, Z. Hu, and F. Zhang, "Multitemporal SAR and polarimetric SAR optimization and classification: Reinterpreting temporal coherence," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5236617.

[12] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512913.

[13] Y. Duan, F. Luo, M. Fu, Y. Niu, and X. Gong, "Classification via structure-preserved hypergraph convolution network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507113.

[14] J. Geng, R. Wang, and W. Jiang, "Polarimetric SAR image classification based on feature enhanced superpixel hypergraph neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5237812.

[15] K. Wu, J. Fan, P. Ye, and M. Zhu, "Hyperspectral image classification using spectral–spatial token enhanced transformer with hash-based positional embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507016.

[16] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[17] G. Liu, Y. Li, Y. Chen, R. Shang, and L. Jiao, "Pol-NAS: A neural architecture search method with feature selection for PolSAR image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9339–9354, 2022.

[18] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[19] W. Liao, R. Bellens, A. Pižurica, S. Gautama, and W. Philips, "Combining feature fusion and decision fusion for classification of hyperspectral and LiDAR data," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1241–1244.

[20] B. Bigdeli, F. Samadzadegan, and P. Reinartz, "Fusion of hyperspectral and LiDAR data using decision template-based fuzzy multiple classifier system," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 38, pp. 309–320, Jun. 2015.

[21] S. Jia et al., "Multiple feature-based superpixel-level decision fusion for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1437–1452, Feb. 2021.

[22] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[23] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[24] H.-C. Li, W.-S. Hu, W. Li, J. Li, Q. Du, and A. Plaza, "A3 CLNN: Spatial, spectral and multiscale attention ConvLSTM neural network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 747–761, Feb. 2022.

[25] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714.

[26] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[27] H. Bao, L. Dong, S. Piao, and F. Wei, "BEiT: BERT pre-training of image transformers," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–13.

[28] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.

[29] R. Girdhar, A. El-Nouby, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "OmniMAE: Single model masked pretraining on images and videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10406–10417.

[30] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 197–211.

[31] C. Ge, Q. Du, W. Li, Y. Li, and W. Sun, "Hyperspectral and LiDAR data classification using kernel collaborative representation based residual fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1963–1973, Jun. 2019.

[32] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.

[33] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^{3}net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.

[34] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[35] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607.

[36] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9630–9640.

[37] C. Liu, H. Sun, Y. Xu, and G. Kuang, "Multi-source remote sensing pretraining based on contrastive self-supervised learning," *Remote Sens.*, vol. 14, no. 18, p. 4632, Sep. 2022.

[38] A. Baevski, W.-N. Hsu, Q. Xu, A. Babu, J. Gu, and M. Auli, "Data2vec: A general framework for self-supervised learning in speech, vision and language," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 1298–1312.

[39] Z. Feng, L. Song, S. Yang, X. Zhang, and L. Jiao, "Cross-modal contrastive learning for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517713.

[40] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–15.

[41] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–12.

[42] D. Ye, Z. Ni, H. Wang, J. Zhang, S. Wang, and S. Kwong, "CSformer: Bridging convolution and transformer for compressive sensing," *IEEE Trans. Image Process.*, vol. 32, pp. 2827–2842, 2023.

[43] H. Zhang et al., "Knowledge amalgamation for object detection with transformers," *IEEE Trans. Image Process.*, vol. 32, pp. 2093–2106, 2023.

[44] Z. Chang, Z. Feng, S. Yang, and Q. Gao, "AFT: Adaptive fusion transformer for visible and infrared images," *IEEE Trans. Image Process.*, vol. 32, pp. 2077–2092, 2023.

[45] T. You, C. Wu, Y. Bai, D. Wang, H. Ge, and Y. Li, "HMF-former: Spatio-spectral transformer for hyperspectral and multispectral image fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[46] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:32:57 UTC from IEEE Xplore. Restrictions apply.

[47] X. Tang, Y. Wang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Interacting-enhancing feature transformer for cross-modal remote-sensing image and text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611715.
[48] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.
[49] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.
[50] N. Park and S. Kim, "How do vision transformers work?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–14.
[51] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.
[52] A. Okujeni, S. van der Linden, and P. Hostert, "Berlin-urban-gradient dataset 2009-an EnMAP preparatory flight campaign," GFZ Data Services, 2016.
[53] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 416–425.
[54] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6504205.
[55] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 8057–8070, Oct. 2023.
[56] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[57] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.



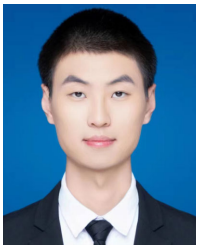
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